综合练习

人员

李瑞涵、田心一、柳力玮、赵书梵、刘宸熙、初锦阳、刘祺、明宗亮、苑钊、杨咏丞、纪博涵 到课, 高健桓 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4524

王向东老师周日三点半C++分组背包

刷新

#	用户名	姓名	编程分	时间	A (11 / 12) (11 / 12)	B (11 / 12) (11 / 12)	C (11 / 22) (11 / 22)	D (9 / 29) (9 / 29)	E (2 / 3) (2 / 3)	F (2 / 2) (2 / 2)
1	zhaoshufan	赵书梵	6	7:2:59	0:8:8	0:29:32	0:48:17	(-1) 1:4:17	1:43:32	2:29:13
2	liuliwei	柳力玮	6	60:54:49	0:18:26	0:43:54	(-1) 1:0:43	(-1) 1:23:6	(-1) 28:6:31	28:22:9
3	tianxinyi	田心一	4	4:22:30	0:23:41	0:45:30	1:20:57	(-1) 1:32:22		
4	jiangshuzhang	蒋叔璋	4	5:11:31	0:26:2	1:0:46	(-2) 1:28:42	1:36:1		
5	liuqi	刘祺	4	5:53:36	0:21:6	0:54:29	1:33:5	(-3) 2:4:56		
6	mingzongliang	明宗亮	4	6:11:34	0:32:1	0:51:28	1:32:16	(-4) 1:55:49		
7	liuchenxi	刘宸熙	4	6:17:32	0:36:52	1:26:30	1:37:16	(-2) 1:56:54		
8	yuanzhao	苑钊	4	7:20:48	(-1) 0:33:18	(-1) 1:11:41	1:35:43	(-4) 2:0:6		
9	liruihan	李瑞涵	4	8:34:26	0:29:36	0:59:15	(-7) 1:59:2	(-2) 2:6:33		
10	chujinyang	初锦阳	3	3:13:30	0:25:27	0:47:6	(-1) 1:40:57			
11	yangyongcheng	杨咏丞	3	430:54:23	140:15:46	150:19:57	140:18:40	(-2)		

本周作业

https://cppoj.kids123code.com/contest/4676 (课上讲了 A ~ C 题, 课后作业是 C 题必做, D 题选做)

课堂表现

今天的题目比较难, 前两题涉及维护倍数, 后两题涉及前缀和维护dp, 同学们做起来难很正常, 继续坚持就可以了。

课堂内容

Max Multiple

f[i][j][k]: 考虑前 i 个数, 选了 j 个, 和模 D 为 k 时, 最大和是多少

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 100 + 5;
```

```
int w[maxn], f[maxn][maxn][maxn];
// f[i][j][k]: 考虑前 i 个数, 选了 j 个, 和模 D 为 k 时, 最大和是多少

signed main()
{
  int n, K, D; cin >> n >> K >> D;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  memset(f, -0x3f, sizeof(f)); f[0][0][0] = 0;
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j <= K; ++j) {
      for (int k = 0; k <= D-1; ++k) {
        f[i][j][k] = f[i-1][j][k];
        if (j) f[i][j][k] = max(f[i][j][k], f[i-1][j-1][(k-w[i]%D+D)%D] + w[i]);
      }
    }
  }

  if (f[n][K][0] >= 0) cout << f[n][K][0] << endl;
  else cout << -1 << endl;
  return 0;
}
```

I Hate Non-integer Number

f[i][j][k]: 考虑前 i 个数, 选了 j 个, 总和模 C 为 k 时有多少方案

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 100 + 5;
const int mod = 998244353;
int w[maxn], f[maxn][maxn][maxn];
// f[i][j][k]: 考虑前 i 个数, 选了 j 个, 总和模 C 为 k 时有多少方案
int n;

int solve(int C) {
  memset(f, 0, sizeof(f));

  f[0][0][0] = 1;
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j <= i; ++j) {
      for (int k = 0; k < C; ++k) {
        f[i][j][k] = f[i-1][j][k];

        if (j) {
          f[i][j][k] += f[i-1][j-1][(k-w[i]%C+C)%C];
          f[i][j][k] %= mod;
        }
```

```
            }
        }
    }

    return f[n][C][0];
}

signed main()
{
    cin >> n;
    for (int i = 1; i <= n; ++i) cin >> w[i];

    int res = 0;
    for (int i = 1; i <= n; ++i) res += solve(i), res %= mod;
    cout << res << endl;
    return 0;
}
```

Distance Sequence

f[i][j]: 考虑前 i 个数, 第 i 个数为 j 时, 有多少方案

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int N = 1000 + 5, M = 5000 + 5;
const int mod = 998244353;
int f[N][M], p[N][M];

int get_sum(int id, int l, int r) {
    if (l > r) return 0;
    return (p[id][r]-p[id][l-1]+mod)%mod;
}

signed main()
{
    int n, m, K; cin >> n >> m >> K;
    for (int i = 1; i <= n; ++i) {
        for (int j = 1; j <= m; ++j) {
            if (i == 1) f[i][j] = 1;
            else if (K) f[i][j] = (get_sum(i-1,1,j-K) + get_sum(i-1,j+K,m)) % mod;
            else f[i][j] = get_sum(i-1,1,m);
        }
        for (int j = 1; j <= m; ++j) p[i][j] = (p[i][j-1]+f[i][j]) % mod;
    }

    int res = 0;
    for (int i = 1; i <= m; ++i) res = (res + f[n][i]) % mod;
    cout << res << endl;
```

```
  return 0;
}
```

Leaping Tak

f[i]: 走到 i 有多少方案

用前缀和优化转移

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int N = 2e5 + 5, M = 10 + 5;
const int mod = 998244353;
struct node {
  int l, r;
} w[M];
int f[N], p[N];

int get_sum(int l, int r) {
  if (r <= 0) return 0;
  if (l <= 0) return p[r];

  return (p[r] - p[l-1] + mod) % mod;
}

signed main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= m; ++i) cin >> w[i].l >> w[i].r;
  f[1] = p[1] = 1;

  for (int i = 2; i <= n; ++i) {
    for (int j = 1; j <= m; ++j) {
      f[i] = (f[i] + get_sum(i-w[j].r, i-w[j].l)) % mod;
      p[i] = (p[i-1] + f[i]) % mod;
    }
  }

  cout << f[n] << endl;
  return 0;
}
```